DATASCI 406 FA 26, Quiz 3.1

Name	Uniqname	UMID

Each question is worth 1 point. Circle one answer for each.

Question 1 (1 point)

A bootstrap SE for a sample of $n = 30$ observations is computed from $B = 1000$ bootstrap samples. What changes if B is raised to 100,000?

1. The bootstrap SE converges to the true SE of T.
2. Both the Monte Carlo error and the error from using $\hat{F}$ in place of F shrink.
3. The Monte Carlo error shrinks, but the error from using $\hat{F}$ in place of F does not.
4. Nothing, because every one of the 30^{30} possible bootstrap samples is already drawn.

Question 2 (1 point)

When would you use a basic or percentile interval instead of the normal interval, $T \pm z_{1-\alpha/2}\,\widehat{\mathrm{SE}}$?

1. When the bootstrap distribution of T^* is skewed.
2. When B is small, because quantiles need fewer bootstrap samples than an SE.
3. When T is the mean of a large sample.
4. When the bootstrap SE is large.

Question 3 (1 point)

Each of `n` homes has a first-draw lead level `x[k]` and a level after flushing `y[k]`. Which code computes one bootstrap replicate of the mean difference of logs?

1. `mean(log(sample(x, replace = TRUE)) - log(sample(y, replace = TRUE)))`
2. `i <- sample(n); mean(log(x[i]) - log(y[i]))`
3. `i <- sample(n, replace = TRUE); mean(log(x[i])) - mean(log(y))`
4. `i <- sample(n, replace = TRUE); mean(log(x[i]) - log(y[i]))`

Question 4 (1 point)

For which of these statistics would the bootstrap **not** give a good estimate of the sampling distribution?

1. The sample correlation.
2. The sample maximum.
3. The sample median.
4. The mean difference of paired observations.